

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING, KALAVAKKAM
(An Autonomous Institution, Affiliated to Anna University, Chennai)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III Year CSE (V Semester)
Academic Year 2026-27
UCS3513 SYSTEM DESIGN LABORATORY

Lab Exercise 3- Improve URL Shortener Performance Using Redis Caching

Objective

To integrate Redis into the URL Shortener application developed in Experiment 2 and evaluate its impact on application performance by comparing URL retrieval with and without caching.

Problem Statement

Enhance the URL Shortener application by implementing Redis as an in-memory cache for URL mappings.

1. Configure Redis with the Spring Boot application.
2. Cache URL mappings using the **short code** as the cache key.
3. Implement a **cache-aside strategy**:
 - Check Redis for the requested short code.
 - If found (**Cache Hit**), retrieve the long URL from Redis.
 - Otherwise (**Cache Miss**), retrieve the long URL from the database, store it in Redis, and return the result.
4. Compare the application's performance before and after integrating Redis.
5. Analyze the impact of Redis on response time and database access.

Tasks

- Configure and integrate Redis with the existing Spring Boot application.
- Modify the **GET /{shortCode}** API to retrieve URL mappings from Redis before accessing the database.
- Implement cache update logic on a cache miss.
- Test the application using Postman.
- Measure the response time of URL retrieval:
 - **Without Redis**
 - **With Redis**
- Compare the results and summarize the performance improvement.

Expected Learning Outcomes

After completing this experiment, students will be able to:

- Integrate Redis with a Spring Boot application.
- Implement a cache-aside caching strategy.
- Understand the role of caching in improving application performance.
- Compare the performance of database-based retrieval and Redis-based retrieval.
- Analyze cache hit and cache miss scenarios.